Nils Degardin 12/9/24

Algebra Based Physics - Midterm 2

Unit 4

1. $\dfrac{60\ s}{1.33\ s/\text{rotation}} \approx 45.11$ rotations / minute

$\dfrac{2\pi}{1.33} \approx 4.72$ radians / s

B 45 rpm, 4.7 radians sec^{-1}

2. $a_c = r\omega^2$

$\omega = 4.7$ radians s^{-1}

$r = .1\ m$

$.1 \times 4.7 = 2.2$

$m \left(\frac{rad}{s}\right)^2 \left(\frac{m}{s^2}\right)$

D $2.2\ m/s^2$

3. $a_c = r\omega^2$

$a'_c = 100 \cdot a_c$

$r\omega^2_{new} = 100 \cdot r\omega^2$

$\omega^2_{new} = 100 \cdot \omega^2$

$\omega_{new} = 10 \cdot \omega$

D $\omega \to 10\omega$

4. $\omega = 200\,rpm \cdot \dfrac{2\pi\ rad}{60\ s}$

$\omega = \dfrac{200 \cdot 2\pi}{60} \approx 20.94\ rad/s$

$a_c = 0.12\,(20.94)^2$

$\approx 0.12 \cdot 438.53 \approx 52.62\ m/s^2$

$F_c = 0.01 \cdot 52.62 \approx$ B $0.53\ N$

5. $v = 1.40 \cdot 30$

$v = 42\ m/s$ C

Unit 5

1. $W = (40)(5) + (-30)(0) = 200\ N$

$|\vec{F}| = \sqrt{(40)^2 + (30)^2} = \sqrt{1600 + 900} = \sqrt{2500} = 50\ N$

$|\vec{\Delta x}| = \sqrt{(5)^2} = 5\ m$

$\vec{F} \cdot \vec{\Delta x} = 200\ N$

$\cos\theta = \frac{200}{(50)(5)} = \frac{200}{250} = 0.8 \qquad \theta = \cos^{-1}(0.8) \approx 37°$

D 200 N·m, 37 degrees

2. a) $W = F \cdot d \qquad W = 1000 \cdot 0.1 = 100\ J$

b) $\Delta KE = W \qquad \Delta KE = 100\ J$

c) $100 = \frac{1}{2} \cdot 0.15 \cdot v^2 f$

$v^2 f = \frac{100 \cdot 2}{0.15} = \frac{200}{0.15} \approx 1333.33$

$v f \approx \sqrt{1333.33} \approx 36.5\ m/s$

d) $R = \frac{v^2 f \sin(2\theta)}{g}$

$R = \frac{(36.5^2) \cdot 1}{9.8} \qquad R = \frac{1332.25}{9.8} \approx 136\ m$

3. a) $F_n = 40 \cdot 9.8 = 392\ N \qquad F_k = 0.05 \cdot 392 = 19.6\ N$

$W = 19.6 \cdot 40 = 784\ J$

b) $\frac{784}{30} \approx 26.1\ W$

4. Power (kW) $= \frac{3W}{1000} = 0.003\ kW$

Time (hours) $365 \cdot 24 = 8760$ hours

Energy (kWh) $= 0.003 \cdot 8760 = 26.28\ kWh$

Rate $= \frac{9}{100} = 0.09\ \$/kWh$

Cost $= 26.28 \cdot 0.09 = 2.37\ \$$ D

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Unit 6

1. $P_{initial} = (20 \times 10^{-25}) \cdot 350 + (20 \times 10^{-25}) \cdot (-350)$

$= 7 \times 10^{-23} - 7 \times 10^{-23} = 0$

$M v_f = 0 \quad v_f = 0 \text{ m/s}$ C

2. $p = (20 \times 10^{-25}) \cdot 350 = 7 \times 10^{-23} \text{ m/s}$

$p_r = \sqrt{2p^2 + \sqrt{2}p^2} \quad p_r = p\sqrt{2+\sqrt{2}}$

$p_r = (7 \times 10^{-23}) \cdot \sqrt{2+\sqrt{2}} \approx 1.293 \times 10^{-22} \text{ m/s}$

$v_f = \dfrac{1.293 \times 10^{-22}}{40 \times 10^{-25}}$

$v_f = \dfrac{1.293}{40} \times 10^{2} = 3.23 \text{ m/s}$

3. a. Impulse $= 4000 \cdot 0.2 = 800 \text{ N}$

b. $800 = 200(v_f - 2.8)$

$4 = v_f - 2.8$

$v_f = 4 + 2.8 = 6.8 \text{ m/s}$

4. D totally elastic
- kinematic energy is conserved fully
- momentum has stayed the same
- no energy was lost

5. a) $(30{,}000 \text{ kg})(0.850 \text{ m/s}) + (110{,}000 \text{ kg})$

$= (30{,}000) + (110{,}000) v_f$

$\dfrac{25{,}500 \text{ kg} \cdot \text{m/s}}{140{,}000 \text{ kg}} \approx 0.182 \text{ m/s}$

b) $KE_{initial} = \frac{1}{2}(30{,}000)(0.850)$

$= \frac{1}{2}(30\,000)(0.7225) = 10{,}837.5 \text{ J}$

$KE_{final} = \frac{1}{2}(140000)(0.182)$

$= \frac{1}{2}(140\,000\,(0.0331) = 2{,}317 \text{ J}$

$10{,}837.5 \text{ J} - 2{,}317 \text{ J} = 8{,}520.5 \text{ J}$

Unit 7

1. $F_{bricks} = m_{bricks} \cdot g = 40 \cdot 9.8 = 392\ N$

$\tau_{bricks} = F_{bricks} \cdot r_{bricks} = 392 \cdot 3 = 1176\ N$

counter clockwise rotation

$F_{concrete} = m_{concrete} \cdot g = 60 \cdot 9.8 = 588\ N$

$\tau_{concrete} = F_{concrete} \cdot r_{concrete} = 588 \cdot 2 = 1176\ N$

equal in magnitudes and opposite directions so it will remain motionless because they cancel each other out

A

2. Net torque = 0 $\qquad 20 \cdot 9.81 = 196.2 \Rightarrow F_1 + F_2 = 196.2$

Net force = 0

$F_1 \cdot 1.5 = W \cdot 0.5 \qquad W = 196$

$F_1 \cdot 1.5 = 196 \cdot 0.5$

$\frac{F_1 \cdot 1.5}{1.5} = \frac{98}{1.5} \qquad F_1 = 65.33\ N$ at the end

$196 - 65.33 = 130.67$

$F_2 = 130.67$ 0.5 m from the end

Unit 8

1. a) $I = \frac{1}{2} \times 1\,kg \times (0.08\,m)^2$

$\frac{1}{2} \times 1 \times 0.0064 = 0.0032\ kg \cdot m^2$

$I = 32 \times 10^{-4}\ kg\ m^2$

b) $0.0032 \times 2 = 0.0064\ N \cdot m \quad \tau = 64 \times 10^{-4}\ N \cdot m$

2. a) $r = -20i + 20j\ cm \rightarrow -.20i + .20j\ m$

$F = 12\hat{i}\ N \qquad (-.20\hat{i} + .20\hat{j}) \cdot 12\hat{i} \qquad 2.4\ k\ Nm$

b) $\tau = r \cdot F \rightarrow -\tau = r \cdot F \qquad F = F_x\hat{i} + F_y\hat{j}$

$-\tau = 2.4\ k\ Nm \qquad 2.4\hat{k} - (-.20\hat{i} + .20\hat{j})(F_x\hat{i} + F_y\hat{j})$

$r = -.20\hat{i} + .20\hat{j}\ m \qquad -.20\hat{i}(F_x\hat{i} + F_y\hat{j}) + .20(F_x\hat{i} + F_y\hat{j})$

$.20 F_x(-\hat{k})$

$-.20 F = x(+k) = \frac{2.4\hat{k}}{-.20} = 12\ N$